

12 (long)

(a) Distinguish a thermoplastic from a thermoset. Give two examples of each. Sketch the variation of Young's modulus with temperature for

- (i) an amorphous thermoplastic;
- (ii) a thermoset.

Label critical temperatures and account for transitions in behaviour.

[10]

(b) The viscoelastic response of polymers can be modelled by a system of linear elastic springs and viscous dashpots. Write down the governing equations for the stress for each of the two components. Without carrying out any calculations, sketch strain versus time for the case of an imposed step in stress when a spring is connected with a dashpot (i) in series and (ii) in parallel. Explain why the plots take the forms you have sketched.

[8]

(c) The deformation response of a polymer is to be represented by a linear dashpot of viscosity η in series with a linear spring of Young's modulus E_1 , as shown in Fig. 7(a). Derive the governing differential equation relating the stress to the strain. Hence find the harmonic response.

[8]

(d) To simulate the deformation response of another polymer, an additional spring of Young's modulus E_2 is now loaded in parallel with the system from part (c), as shown in Fig. 7(b). Use the "electrical circuit method" to write down the harmonic response.

[4]

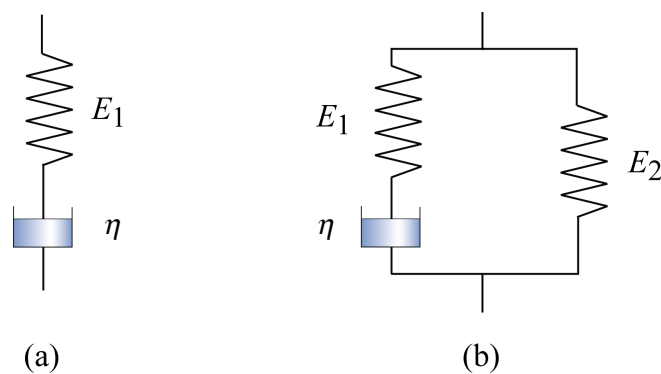


Fig. 7

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Answers

1. $\sigma_t = \frac{-PL}{4t}, \quad \sigma_c = \frac{-PL}{2t}$

2. $FR \left(\frac{1}{\sqrt{3}} - \frac{1}{2} \right)$

3. $\frac{10FL}{AE}$ downwards.

4. $\tan \psi = \frac{\Delta e}{e_0} \tan \theta$

5. (b) $4.1 \times 10^9 \text{ mm}^4$ (c) 600 kNm

6. (b) $d(x) = \frac{3\omega|x|}{b\sigma_y}$

(c) the two designs intersect at $x = \frac{L}{\sqrt{1+k}}, d(x) = \frac{kL}{\sqrt{1+k}}$ where $k = \frac{3\omega}{b\sigma_y}$

7. (a) Atomic density BCC: $91.1 \times 10^{27} \text{ atoms m}^{-3}$, FCC: $93.3 \times 10^{27} \text{ atoms m}^{-3}$

(b) $\frac{\Delta L}{L} \approx -0.79\%$

8. (a) $\sigma_y \approx 213 \text{ MPa}, H = 639 \text{ MPa}$. (b) $\varepsilon_n = -0.20$ i.e. 20% reduction, $\frac{F_c}{F_t} = 1.56$.

11. (a) (i) $\sigma_{\max} = 376 \text{ MPa}, \Delta\sigma = 338.4 \text{ MPa}$;

(ii) $a_o = 35 \text{ }\mu\text{m}$, initial crack length $2 \times a_o$;

(b) $\sigma = 427 \text{ MPa}$

12. (c) $\frac{\hat{\varepsilon}}{\hat{\sigma}} = \frac{\left(i\omega + \frac{E_1}{\eta} \right)}{i\omega E_1}$ (d) $\frac{\hat{\varepsilon}}{\hat{\sigma}} = \frac{E_1 + i\omega\eta}{E_2(E_1 + i\omega\eta) + E_1 i\omega\eta}$